



Platinum

PYTHON

COURSE

COURSE HIGHLIGHTS

DESKTOP AND WEB APPLICATIONS
AI & MACHINE LEARNING & DSA
WEB SCRAPPING

6-7 MONTHS
Duration



Why Python ?



Extensive support libraries



Portable and Interactive



User-friendly data structures



Ideal for prototypes

PYTHON

Python is a high-level, general-purpose and a very popular programming language. Python programming language (latest Python 3) is being used in web development, Machine Learning applications, along with all cutting edge technology in Software Industry.

Python is relatively simple, so it's easy to learn since it requires a unique syntax that focuses on readability. Developers can read and translate Python code much easier than other languages.



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77107-88822 (Hamirpur)



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**BOOK YOUR
SEAT**

PYTHON PLATINUM SYLLABUS

WEEK 1 - 5

INTRO TO PYTHON

- WHAT IS PYTHON AND HISTORY OF PYTHON?
- UNIQUE FEATURES OF PYTHON
- PYTHON-2 AND PYTHON-3 DIFFERENCES
- INSTALL PYTHON AND ENVIRONMENT SETUP
- FIRST PYTHON PROGRAM
- PYTHON IDENTIFIERS, KEYWORDS AND INDENTATION
- COMMENTS AND DOCUMENT INTERLUDE IN PYTHON
- COMMAND LINE ARGUMENTS
- GETTING USER INPUT
- PYTHON DATA TYPES
- WHAT ARE VARIABLES?
- PYTHON CORE OBJECTS AND FUNCTIONS
- NUMBER AND MATHS
- WEEK 1 ASSIGNMENTS

CONTROL STATEMENTS

- IF-ELSE
- IF-ELIF-ELSE
- WHILE LOOP
- FOR LOOP
- BREAK
- CONTINUE
- ASSERT
- PASS
- RETURN

LIST, RANGES & TUPLES IN PYTHON

- INTRODUCTION
- LISTS IN PYTHON
- MORE ABOUT LISTS
- UNDERSTANDING ITERATORS
- GENERATORS, COMPREHENSIONS AND LAMBDA EXPRESSIONS
- UNDERSTANDING AND USING RANGES
- ORDERED SETS WITH TUPLES

WEEK 6 - 7

PYTHON DICTIONARIES AND SETS

- INTRODUCTION TO THE SECTION
- PYTHON DICTIONARIES
- MORE ON DICTIONARIES
- SETS
- PYTHON SETS EXAMPLES

INPUT AND OUTPUT IN PYTHON

- READING AND WRITING TEXT FILES
- WRITING TEXT FILES

- APPENDING TO FILES AND CHALLENGE

PYTHON BUILT IN FUNCTION

- PYTHON USER DEFINED FUNCTIONS
- PYTHON PACKAGES FUNCTIONS
- DEFINING AND CALLING FUNCTION
- THE ANONYMOUS FUNCTIONS
- LOOPS AND STATEMENT IN PYTHON
- PYTHON MODULES & PACKAGES

EVALUATION -1

PYTHON OBJECT ORIENTED

- OVERVIEW OF OOP
- THE SELF VARIABLE
- CONSTRUCTOR
- TYPES OF VARIABLES
- NAMESPACES
- CREATING CLASSES AND OBJECTS
- INHERITANCE
- TYPES OF METHODS
- INSTANCE METHODS
- STATIC METHODS
- CLASS METHODS
- ACCESSING ATTRIBUTES
- BUILT-IN CLASS ATTRIBUTES
- DESTROYING OBJECTS
- ABSTRACT CLASSES AND INTERFACES
- ABSTRACT METHODS AND ABSTRACT CLASS
- INTERFACE IN PYTHON
- ABSTRACT CLASSES AND INTERFACES

EXCEPTIONS

- ERRORS IN PYTHON
- COMPILE-TIME ERRORS
- RUNTIME ERRORS
- LOGICAL ERRORS
- WHAT IS EXCEPTION?
- HANDLING AN EXCEPTION
- TRY....EXCEPT...ELSE
- TRY-FINALLY CLAUSE
- ARGUMENT OF AN EXCEPTION
- PYTHON STANDARD EXCEPTIONS

PYTHON REGULAR EXPRESSIONS

- WHAT ARE REGULAR EXPRESSIONS?
- THE MATCH FUNCTION
- THE SEARCH FUNCTION
- MATCHING VS SEARCHING
- SEARCH AND REPLACE
- EXTENDED REGULAR EXPRESSIONS
- WILDCARD

MULTITHREADED PROGRAMMING

- WHAT IS MULTITHREADING?
- DIFFERENCE BETWEEN A PROCESS AND THREAD
- CONCURRENT PROGRAMMING AND GIL
- USES OF THREAD
- STARTING A NEW THREAD
- THE THREADING MODULE
- THREAD SYNCHRONIZATION

USING DATABASES IN PYTHON

- PYTHON MYSQL DATABASE ACCESS
- INSTALL THE MYSQLDB AND OTHER PACKAGES
- CREATE DATABASE CONNECTION
- CREATE, INSERT, READ OPERATION
- DML AND DDL OPERATION WITH DATABASES

EVALUATION -2

WEEK 11

GUI USING TKINTER

- TKINTER BASICS
- TKINTER PROGRAMMING
- GUI PROGRAMMING
- TKINTER WIDGETS
- TKINTER LISTBOX
- TKINTER BUTTON
- TKINTER CHECKBUTTON
- TKINTER ENTRY
- TKINTER FRAME
- TKINTER MENUBUTTON
- TKINTER MENU WIDGET
- TKINTER MESSAGE WIDGET
- TKINTER MESSAGEBOX WIDGET
- TKINTER RADIOBUTTON WIDGET
- TKINTER SCALE WIDGET
- TKINTER SCROLLBAR WIDGET
- TKINTER TEXT
- TKINTER TOPLEVEL WIDGET
- TKINTER SPINBOX
- TKINTER PANEDWINDOW
- TKINTER LABELFRAME
- TKINTER TKMESSAGEBOX
- TKINTER PANEDWINDOW WIDGET
- TKINTER LABELFRAME WIDGET

GEOMETRY MANAGEMENT

- TKINTER PACK() METHOD
- TKINTER GRID() METHOD
- TKINTER PLACE() METHOD

PROJECT WORK

- CALCULATRO APP USING TKINTER
- AGE CALCULATOR APP
- PERCENTILE CALCULATOR
- COMPOUND INTEREST CALCULATOR APP

WEEK 12

PROJECT WORK

- CALCULATRO APP USING TKINTER
- AGE CALCULATOR APP
- PERCENTILE CALCULATOR
- COMPOUND INTEREST CALCULATOR APP

EVALUATION -3

FLASK

- INTRODUCTION AND OVERVIEW
- ENVIRONMENT
- APPLICATION
- ROUTING
- VARIABLES RULES
- URL BUILDING
- HTTP METHODS
- TEMPLATES
- STATIC FILES
- REQUEST OBJECT
- SENDING FORM DATA TO TEMPLATE
- COOKIES
- SESSIONS
- REDIRECT AND ERRORS
- MESSAGE FLASHING
- FILE UPLOADING
- EXTENSIONS
- MAIL
- WTF
- SQLITE
- SQLALCHEMY
- DEPLOYMENT
- CGI

PROJECT WORK

EVALUATION - 4

DJANGO

- COVERING BASICS AND OVERVIEW
- ENVIRONMENT • CREATING A PROJECT
- LIFE CYCLE OF APPS
- ADMIN INTERFACE
- CREATING VIEWS
- URL MAPPING
- TEMPLATE SYSTEM
- MODELS
- PAGE REDIRECTION
- SENDING EMAILS
- GENERIC VIEWS
- FORM PROCESSING
- FILE UPLOADING
- APACHE SETUP
- COOKIES HANDLING
- SESSIONS
- CACHING
- COMMENTS
- RSS
- AJAX

RESTFUL API

- INTRODUCTION
- ENVIRONMENT SETUP
- RESOURCES
- MESSAGES
- ADDRESSING
- METHODS
- STATELESSNESS
- CACHING
- SECURITY
- JAVA (JAX- RS)

PYTHON WEB SCRAPING

- REQUESTS MODULE
- INSTALLATION
- MAKING A REQUEST
- RESPONSE OBJECT
- BEAUTIFULSOUP LIBRARY
- INSTALLATION
- INSPECTING WEBSITE
- PARSING THE HTML
- FINDING ELEMENTS
- FINDING ELEMENTS BY CLASS
- FINDING ELEMENTS BY ID
- EXTRACTING TEXT FROM THE TAGS

MACHINE LEARNING & DATA SCIENCE

- GETTING STARTED WITH MACHINE LEARNING
- AN INTRODUCTION TO MACHINE LEARNING
- WHAT IS MACHINE LEARNING ?
- INTRODUCTION TO DATA IN MACHINE LEARNING
- DEMYSTIFYING MACHINE LEARNING
- ML – APPLICATIONS
- BEST PYTHON LIBRARIES FOR MACHINE LEARNING
- ARTIFICIAL INTELLIGENCE | AN INTRODUCTION
- MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE
- DIFFERENCE BETWEEN MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

DATA AND IT'S PROCESSING

- INTRODUCTION TO DATA IN MACHINE LEARNING
- UNDERSTANDING DATA PROCESSING
- PYTHON | CREATE TEST DATASETS USING SKLEARN
- PYTHON | GENERATE TEST DATASETS FOR MACHINE LEARNING
- PYTHON | DATA PREPROCESSING IN PYTHON
- DATA CLEANING

SUPERVISED LEARNING

- GETTING STARTED WITH CLASSIFICATION
- BASIC CONCEPT OF CLASSIFICATION
- TYPES OF REGRESSION TECHNIQUES
- CLASSIFICATION VS REGRESSION
- ML | TYPES OF LEARNING – SUPERVISED LEARNING
- MULTICLASS CLASSIFICATION USING SCIKIT-LEARN

GRADIENT DESCENT

- GRADIENT DESCENT ALGORITHM AND ITS VARIANTS
- STOCHASTIC GRADIENT DESCENT (SGD)
- MINI-BATCH GRADIENT DESCENT WITH PYTHON

LINEAR REGRESSION

- INTRODUCTION TO LINEAR REGRESSION
- GRADIENT DESCENT IN LINEAR REGRESSION
- MATHEMATICAL EXPLANATION FOR LINEAR REGRESSION WORKING
- NORMAL EQUATION IN LINEAR REGRESSION
- LINEAR REGRESSION (PYTHON IMPLEMENTATION)
- UNIVARIATE LINEAR REGRESSION IN PYTHON
- MULTIPLE LINEAR REGRESSION USING PYTHON
- MULTIPLE LINEAR REGRESSION USING R
- LOCALLY WEIGHTED LINEAR REGRESSION
- GENERALIZED LINEAR MODELS

LOGISTIC REGRESSION

- UNDERSTANDING LOGISTIC REGRESSION
- WHY LOGISTIC REGRESSION IN CLASSIFICATION ?
- LOGISTIC REGRESSION USING PYTHON
- COST FUNCTION IN LOGISTIC REGRESSION
- LOGISTIC REGRESSION USING TENSORFLOW

UNSUPERVISED LEARNING

- ML | TYPES OF LEARNING – UNSUPERVISED LEARNING
- SUPERVISED AND UNSUPERVISED LEARNING
- CLUSTERING IN MACHINE LEARNING
- DIFFERENT TYPES OF CLUSTERING ALGORITHM
- K MEANS CLUSTERING – INTRODUCTION
- ELBOW METHOD FOR OPTIMAL VALUE OF K IN KMEANS
- RANDOM INITIALIZATION TRAP IN K-MEANS
- ML | K-MEANS++ ALGORITHM
- ANALYSIS OF TEST DATA USING K-MEANS CLUSTERING IN PYTHON

REINFORCEMENT LEARNING

- REINFORCEMENT LEARNING
- REINFORCEMENT LEARNING ALGORITHM : PYTHON IMPLEMENTATION USING Q-LEARNING
- INTRODUCTION TO THOMPSON SAMPLING
- GENETIC ALGORITHM FOR REINFORCEMENT LEARNING
- SARSA REINFORCEMENT LEARNING
- Q-LEARNING IN PYTHON

EVALUATION - 5



GITHUB– PART 1

- HISTORY OF GIT
- DESIGN PRINCIPLES
- DISTRIBUTED VERSION CONTROL
- ACCOUNT SETUP
- INSTALLING SOURCETREE GIT GUI
- COMMON GIT COMMANDS
- CONFIGURING GIT
- CREATING REPOSITORIES
- CREATING A COMMIT

GITHUB– PART 2

- VISUALISING BRANCHES
- BRANCH NAMING CONVENTIONS
- CREATING A NEW BRANCH
- HANDLING MERGE CONFLICTS
- CREATING A MERGE REQUEST
- ACCEPTING A MERGE REQUEST
- REJECTING A MERGE REQUEST

CODE REVIEW + DEPLOYMENT

HOSTING + DOMAIN + SSL + AWS + FIREBASE

30 DAYS LIVE PROJECT INTERNSHIP



Grow Your Skills and Get Hired as a Python Developer



Millions of Python Jobs are Available in 2023



Immediate Hiring



High Salary Packages



Yearly Salary

80%

\$ 160,000.98



The Trainer who teaches Python has immense knowledge about the subject and the training is friendly you can clarify your doubts in the class room itself,

Sushil Pandey



I joined Excellence Technology for python classes. All trainers are very good, experienced with detailed Knowledge about the subject.

Shweta



What Our Student Says

Ready to Supercharge Your Career?

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